

Notation: The index set for the M subsystems is represented by $\mathcal{I}_M = \{1, 2, \dots, M\}$. The sets of nonnegative and positive integers are denoted by $\mathbb{N}$ and $\mathbb{N}^*$, respectively. The symbol $\doteq$ stands for a first-order approximation, as in [2].

1 Statement of the problem

Consider a switched affine system with dynamics described by a model of the form [1]

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)} \quad (1)$$

where $t \in \mathbb{R}_+$ is the continuous time variable, $x(t) \in \mathbb{R}^n$ is the state vector, $\sigma : \mathbb{R}_+ \rightarrow \mathcal{I}_M$ is a piecewise constant and right-continuous switching function, and $A_m \in \mathbb{R}^{n \times n}$, $b_m \in \mathbb{R}^n$ are the model parameters of the m th subsystem.

Let $\bar{x}(t)$ denote a target state trajectory associated to an admissible switching profile $\bar{\sigma}(t)$ of the form

$$\bar{\sigma}(t) = \bar{\sigma}(\bar{t}_{i-1}), \bar{t}_{i-1} \leq t < \bar{t}_i, \forall i \in \mathbb{N}^* \quad (2)$$

where $\bar{t}_i$ is the i th switching instant. Both $\bar{x}(t)$ and $\bar{\sigma}(t)$ are assumed to be periodic, with period T . The number of switchings within one period is denoted by N . The time interval between the switching instants $\bar{t}_{i-1}$ and $\bar{t}_i$ will be indicated as $\Delta\bar{t}_i$, i.e.

$$\Delta\bar{t}_i = \bar{t}_i - \bar{t}_{i-1}, i \in \mathbb{N}^* \quad (3)$$

The problem addressed herein consists in determining perturbations δt_i in the switching time intervals $\Delta\bar{t}_i$ in order to drive the state from an initial point $x(0)$ to the target trajectory. In what follows, the perturbed switching time intervals will be denoted by

$$\Delta t_i = \Delta\bar{t}_i + \delta t_i, i \in \mathbb{N}^* \quad (4)$$

The associated switching function $\sigma(t)$ will be defined as

$$\sigma(t) = \bar{\sigma}(\bar{t}_{i-1}), t_{i-1} \leq t < t_i, \forall i \in \mathbb{N}^* \quad (5)$$

where

$$t_i = t_{i-1} + \Delta t_i, i \in \mathbb{N}^* \quad (6)$$

We shall impose the constraint

$$t_i = \bar{t}_i, \forall i = kN, k \in \mathbb{N} \quad (7)$$

Moreover, it is assumed that the dwell time at the m th subsystem is lower-bounded by a constant $d_m > 0$ i.e. if $\sigma(t_{i-1}) = m$, the time interval $\Delta t_i = t_i - t_{i-1}$ cannot be smaller than d_m .

2 Linearization around the target trajectory

The solution of the state equation (1) starting from an initial condition $x(0)$ can be expressed in a compact manner by using the following augmented state vector [Citar Patino, 2010]:

$$\chi(t) = \begin{bmatrix} x(t) \\ 1 \end{bmatrix} \quad (8)$$

From (1) and (8), it follows that

$$\dot{\chi}(t) = \begin{bmatrix} A_{\sigma(t)}x(t) + b_{\sigma(t)} \\ 0 \end{bmatrix} = \begin{bmatrix} A_{\sigma(t)} & b_{\sigma(t)} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x(t) \\ 1 \end{bmatrix} = \tilde{A}_{\sigma}(t)\chi(t) \quad (9)$$

where

$$\tilde{A}_{\sigma}(t) = \begin{bmatrix} A_{\sigma(t)} & b_{\sigma(t)} \\ 0 & 0 \end{bmatrix} \quad (10)$$

For a given switching function (5), the augmented state at time t_N will be expressed as

$$\chi(t_N) = \exp(\tilde{A}_{\sigma(\bar{t}_{N-1})}\Delta t_N) \exp(\tilde{A}_{\sigma(\bar{t}_{N-2})}\Delta t_{N-1}) \cdots \exp(\tilde{A}_{\sigma(\bar{t}_0)}\Delta t_1)\chi(t_0) \quad (11)$$

where

$$\chi(t_0) = \begin{bmatrix} x(t_0) \\ 1 \end{bmatrix} \quad (12)$$

For brevity of notation, let $F_{i+1} = \tilde{A}_{\sigma(\bar{t}_i)}$, $i = 0, 1, \dots, N-1$. The expression in (11) can then be rewritten as

$$\chi(t_N) = \exp(F_N\Delta t_N) \exp(F_{N-1}\Delta t_{N-1}) \cdots \exp(F_1\Delta t_1)\chi(t_0) \quad (13)$$

Similary, we can write

$$\bar{\chi}(\bar{t}_N) = \bar{\chi}(t_N) = \exp(F_N\Delta \bar{t}_N) \exp(F_{N-1}\Delta \bar{t}_{N-1}) \cdots \exp(F_1\Delta \bar{t}_1)\bar{\chi}(t_0) \quad (14)$$

where

$$\bar{\chi}(t_0) = \begin{bmatrix} \bar{x}(t_0) \\ 1 \end{bmatrix}, \quad \bar{\chi}(t_N) = \begin{bmatrix} \bar{x}(t_N) \\ 1 \end{bmatrix} \quad (15)$$

Now, let

$$e(t) = \chi(t) - \bar{\chi}(t) \quad (16)$$

From (4), (13), (14), and (16), it follows that

$$\begin{aligned} e(t_N) &= \chi(t_N) - \bar{\chi}(t_N) \\ &= \exp(F_N\Delta t_N) \exp(F_{N-1}\Delta t_{N-1}) \cdots \exp(F_1\Delta t_1) \chi(t_0) \\ &\quad - \exp(F_N\Delta \bar{t}_N) \exp(F_{N-1}\Delta \bar{t}_{N-1}) \cdots \exp(F_1\Delta \bar{t}_1) \bar{\chi}(t_0) \\ &= \exp(F_N(\Delta \bar{t}_N + \delta t_N)) \exp(F_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})) \cdots \exp(F_1(\Delta \bar{t}_1 + \delta t_1)) (\bar{\chi}(t_0) + e(t_0)) \\ &\quad - \exp(F_N\Delta \bar{t}_N) \exp(F_{N-1}\Delta \bar{t}_{N-1}) \cdots \exp(F_1\Delta \bar{t}_1) \bar{\chi}(t_0) \end{aligned} \quad (17)$$

By using the following first-order approximation:

$$e^{At_b} \doteq e^{At_a} + Ae^{At_a}(t_b - t_a) \quad (18)$$

it follows from (17) that

$$\begin{aligned}
e(t_N) &\doteq \exp(F_N \Delta \bar{t}_N) \exp(F_{N-1} \Delta \bar{t}_{N-1}) \cdots \exp(F_1 \Delta \bar{t}_1) (\bar{\chi}(t_0) + e(t_0)) \\
&\quad - \exp(F_N \Delta \bar{t}_N) \exp(F_{N-1} \Delta \bar{t}_{N-1}) \cdots \exp(F_1 \Delta \bar{t}_1) \bar{\chi}(t_0) \\
&\quad + [F_N \exp(F_N \Delta \bar{t}_N) \exp(F_{N-1} \Delta \bar{t}_{N-1}) \cdots \exp(F_1 \Delta \bar{t}_1) \delta t_N \\
&\quad + \exp(F_N \Delta \bar{t}_N) F_{N-1} \exp(F_{N-1} \Delta \bar{t}_{N-1}) \cdots \exp(F_1 \Delta \bar{t}_1) \delta t_{N-1} \cdots \\
&\quad + \exp(F_N \Delta \bar{t}_N) \exp(F_{N-1} \Delta \bar{t}_{N-1}) \cdots F_1 \exp(F_1 \Delta \bar{t}_1) \delta t_1] (\bar{\chi}(t_0) + e(t_0)) \\
&\doteq \exp(F_N \Delta \bar{t}_N) \exp(F_{N-1} \Delta \bar{t}_{N-1}) \cdots \exp(F_1 \Delta \bar{t}_1) e(t_0) \\
&\quad + [F_N \exp(F_N \Delta \bar{t}_N) \exp(F_{N-1} \Delta \bar{t}_{N-1}) \cdots \exp(F_1 \Delta \bar{t}_1) \delta t_N \\
&\quad + \exp(F_N \Delta \bar{t}_N) F_{N-1} \exp(F_{N-1} \Delta \bar{t}_{N-1}) \cdots \exp(F_1 \Delta \bar{t}_1) \delta t_{N-1} \cdots \\
&\quad + \exp(F_N \Delta \bar{t}_N) \exp(F_{N-1} \Delta \bar{t}_{N-1}) \cdots F_1 \exp(F_1 \Delta \bar{t}_1) \delta t_1] \bar{\chi}(t_0)
\end{aligned} \tag{19}$$

References

- [1] M. Benmiloud, A. Benalia, M. Djemai, and M. Defoort. On the local stabilization of hybrid limit cycles in switched affine systems. *IEEE Transactions on Automatic Control*, 64(2):841–846, 2019.
- [2] D. E. Kirk. *Optimal Control Theory: An Introduction*. Dover Publications, 2004.